

Endogenous Policy Leadership through Adjustment Costs: A Recursive Monetary–Fiscal Game for Australia

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Abstract

This paper formulates a recursive monetary–fiscal policy game motivated by strategic-investment models. Monetary and fiscal choices are simultaneous in every period. An authority with an instrument-adjustment cost partly commits its future self because its current instrument becomes a payoff-relevant state; the other authority responds to that inherited position. We compare flexible Markov-perfect Nash, monetary persistence only, fiscal persistence only, bilateral persistence, and cooperation under a distinct true social objective. All mandate and adjustment weights are fixed primitives. The original one-period monetary-leader and fiscal-leader Stackelberg games are retained as benchmarks, not mislabelled as two calendar periods. A quarterly Australian calibration produces stable stationary equilibria. Adjustment costs visibly alter policy paths, but the two static leader allocations are separated by only 0.0010 instrument units after the demand shock and 0.0009 after the cost-push shock. The calibrated game therefore contains the strategic-investment mechanism without producing a strongly identified leader. This negative result is informative: persistence is commitment capacity, but leadership additionally requires a sufficiently strong rival response and a material payoff spillover. A forward-looking New Keynesian extension adds a private-expectations conduit. It amplifies the adjustment-state wedge at intermediate degrees of forward-looking behaviour, but not monotonically: under the pure forward-looking endpoint the calibrated adjustment-specific shares remain small. Seven initial-belief searches locate one stable rational-expectations Markov equilibrium.

Keywords: monetary–fiscal coordination; strategic investment; endogenous leadership; Markov-perfect equilibrium; adjustment costs; Australia.

JEL codes: C73, E52, E58, E61, E62.

1 Introduction

The motivating static model is a one-period policy game. In its simultaneous version, monetary and fiscal authorities choose instruments at the same time. In either Stackelberg version, one authority moves first and understands the other’s within-period reaction. Calling this a “two-stage” game describes the order of moves, not two dates.

The desired dynamic extension is different from simply repeating that static game. Policy is chosen in every period, but institutional or implementation frictions make abrupt changes costly. A choice today then affects tomorrow’s feasible incentives. The authority can use its current choice as a strategic investment in a future position, while its rival reacts to the resulting state. If only one authority has this capacity, it may become leader-like. If both have it, their attempts can reinforce, offset, or create what the industrial-organisation literature calls Stackelberg warfare.

This paper makes that mechanism explicit in a linear–quadratic recursive game. The exercise has five regimes:

1. flexible feedback Nash, with no instrument-adjustment cost;
2. cooperation under a fixed true social loss;
3. a monetary adjustment cost only;
4. a fiscal adjustment cost only; and
5. adjustment costs for both authorities.

The last case is a bilateral commitment-capacity game. We use “meta-Nash” only as an informal description of the two authorities’ leadership attempts. It is not literally an outer Nash game in the adjustment coefficients, because those coefficients are primitives. A literal meta-game would require each authority to choose an institution or commitment technology before policy is played.

The formulation is closest in spirit to [Petkov \(2004\)](#), who studies how intertemporal incentives and adjustment costs permit delegated managers to approach leader-like positions, and to [Jun and Vives \(2004\)](#), who compare open-loop and Markov-perfect equilibria in dynamic games with adjustment costs. Their central lesson carries over: durability creates the opportunity to influence future responses, but the direction and strength of the outcome depend on the strategic cross-effects. [Coury and Petkov \(2010\)](#) applies a related dynamic-incentive logic to monetary delegation and credibility.

The Australian result is deliberately reported without forcing the intended story onto the numbers. Unilateral adjustment costs move the shock-period policy vector toward the corresponding static leader benchmark, but the two leader benchmarks are almost the same as one another. Rule-space proximity is therefore weak evidence of leadership. The baseline is best read as a working model and an identification result: it shows exactly what must become stronger for endogenous leadership to be quantitatively important.

2 Economic environment

2.1 A three-equation macro model plus fiscal dynamics

Let x_t be the output gap, π_t the inflation gap, m_t monetary tightening, f_t fiscal expansion, and b_t a public-debt gap. The model is

$$x_t = \rho_x x_{t-1} + \sigma \pi_{t-1} - \sigma m_t + \psi f_t + d_t, \quad (1)$$

$$\pi_t = \rho_\pi \pi_{t-1} + \kappa x_t + u_t, \quad (2)$$

$$b_{t+1} = \rho_b b_t + \chi_m m_t + \chi_f f_t - \chi_\pi \pi_t, \quad (3)$$

$$d_{t+1} = \rho_d d_t + \varepsilon_{t+1}^d, \quad u_{t+1} = \rho_u u_t + \varepsilon_{t+1}^u. \quad (4)$$

Equations (1) and (2) are a backward-looking empirical counterpart of the IS curve and Phillips curve. Equation (3) adds the fiscal state required for monetary–fiscal interaction. There is no Taylor rule: m_t is chosen optimally by the monetary authority. Likewise, there is no exogenous fiscal rule: f_t is the fiscal authority’s control. Thus the familiar three-equation model becomes a strategic system with two explicit policy rules and a debt transition.

The state observed before policy is

$$s_t = (x_{t-1}, \pi_{t-1}, b_t, m_{t-1}, f_{t-1}, d_t, u_t)'. \quad (5)$$

Substitution of (1)–(4) gives

$$s_{t+1} = As_t + B_M m_t + B_F f_t + \varepsilon_{t+1}. \quad (6)$$

The fourth and fifth rows of this transition set next period's inherited instruments equal to m_t and f_t . These rows are the commitment channel.

2.2 Agency mandates and true social welfare

The monetary authority minimizes

$$L_M = E_0 \sum_{t=0}^{\infty} \beta_M^t \{ \omega_{\pi}^M \pi_t^2 + \omega_x^M x_t^2 + \omega_b^M b_{t+1}^2 + r_M m_t^2 + \lambda_M (m_t - m_{t-1})^2 \}. \quad (7)$$

The fiscal authority minimizes

$$L_F = E_0 \sum_{t=0}^{\infty} \beta_F^t \{ \omega_{\pi}^F \pi_t^2 + \omega_x^F x_t^2 + \omega_b^F b_{t+1}^2 + r_F f_t^2 + \lambda_F (f_t - f_{t-1})^2 \}. \quad (8)$$

The central bank is relatively inflation-conservative, $\omega_{\pi}^M / \omega_x^M > \omega_{\pi}^F / \omega_x^F$, while fiscal policy is relatively output-oriented. Each authority pays only the level and adjustment cost of its own instrument. This ownership convention matters. If both authorities were charged for both instruments, the exercise would partly coordinate them by construction.

Cooperation minimizes a separate social loss,

$$L_S = E_0 \sum_{t=0}^{\infty} \beta_S^t \{ \omega_{\pi}^S \pi_t^2 + \omega_x^S x_t^2 + \omega_b^S b_{t+1}^2 + r_M^S m_t^2 + r_F^S f_t^2 + \lambda_M (m_t - m_{t-1})^2 + \lambda_F (f_t - f_{t-1})^2 \}. \quad (9)$$

This is the “true” objective. It is neither the sum of delegated mandates nor an objective whose weights change across game forms. The mandate weights, social weights, and the four instrument-cost parameters are held fixed when the applicable regime is solved. In the main coordination benchmark the instruments are flexible, so $\lambda_M = \lambda_F = 0$; this isolates a change from decentralised to social objectives without simultaneously introducing a new friction.

3 From a static Stackelberg game to recursive strategic investment

3.1 The one-period benchmarks

Write the one-period loss as

$$\ell_j(s, m, f) = \begin{bmatrix} s' & m & f \end{bmatrix} Q_j \begin{bmatrix} s' & m & f \end{bmatrix}', \quad j \in \{M, F\}. \quad (10)$$

The simultaneous static Nash actions solve

$$\ell_m^M(s, m, f) = 0, \quad \ell_f^F(s, m, f) = 0. \quad (11)$$

For monetary leadership, first compute the fiscal response $f = R_F(s, m)$ from $\ell_f^F = 0$, then choose

$$m^L(s) = \arg \min_m \ell_M[s, m, R_F(s, m)]. \quad (12)$$

Fiscal leadership reverses the order. These are exactly the original Stackelberg games and remain useful reference points for the dynamic model.

At the Australian calibration the follower slopes are small:

$$\frac{\partial f}{\partial m} = 0.0128, \quad \frac{\partial m}{\partial f} = 0.0040. \quad (13)$$

The notation in (13) refers to the monetary-leader game first and the fiscal-leader game second. Small slopes make the two Stackelberg points close to the simultaneous point before any dynamic solution is attempted.

3.2 Stationary feedback Nash

Let $V_j(s) = s'P_j s$ be authority j 's continuation value. A stationary Markov-perfect equilibrium has linear rules

$$m_t = K_M s_t, \quad f_t = K_F s_t. \quad (14)$$

Partition Q_j by state and controls. The simultaneous first-order conditions at each state are

$$\begin{bmatrix} Q_{mm}^M + \beta_M B_M' P_M B_M & Q_{mf}^M + \beta_M B_M' P_M B_F \\ Q_{fm}^F + \beta_F B_F' P_F B_M & Q_{ff}^F + \beta_F B_F' P_F B_F \end{bmatrix} \begin{bmatrix} m_t \\ f_t \end{bmatrix} = - \begin{bmatrix} Q_{ms}^M + \beta_M B_M' P_M A \\ Q_{fs}^F + \beta_F B_F' P_F A \end{bmatrix} s_t. \quad (15)$$

Given the resulting K_M, K_F , define

$$A^c = A + B_M K_M + B_F K_F, \quad S = \begin{bmatrix} I \\ K_M \\ K_F \end{bmatrix}. \quad (16)$$

The coupled Riccati updates are

$$P_j = S' Q_j S + \beta_j (A^c)' P_j A^c, \quad j \in \{M, F\}. \quad (17)$$

The Julia code iterates (15)–(17) to a fixed point. It then checks convergence, closed-loop stability, and convergence from six widely dispersed initial value matrices.

3.3 Where the strategic-investment motive enters

If $\lambda_M > 0$, today's m_t is tomorrow's inherited m_t and affects the monetary authority's next choice. Fiscal policy observes the same state and responds through its equilibrium feedback rule. Schematically, the monetary first-order condition contains

$$\underbrace{\beta_M B_M' V_s^M(s_{t+1})}_{\text{continuation value}} = \text{own persistence effects} + \text{macro-state effects} + \underbrace{\text{effect through } K_F}_{\text{strategic investment}}. \quad (18)$$

The last term is absent from an open-loop path choice because a pre-announced future fiscal path does not react to an off-path state. It is present in the Markov game because fiscal policy reoptimizes at every state. This is the same logical distinction emphasised by Jun and Vives (2004).

Adjustment costs are therefore necessary for the lagged-instrument mechanism, but not sufficient for economically large leadership. The current action must materially alter the future state, the rival must react strongly to that state, and that reaction must matter to the first authority's loss. Ordinary instrument smoothing satisfies only the first condition.

4 Analytical parameter ranges

4.1 Static cross-curvature and equilibrium amplification

Before assigning Australian values, consider the local one-period game. Let a_M and a_F be the effects of the two instruments on the vector of macro targets and let W_j be authority j 's target-weight matrix. Ignoring linear terms that shift the intercepts of the reaction functions, define

$$D_M = r_M + a'_M W_M a_M, \quad C_M = a'_M W_M a_F, \quad (19)$$

$$D_F = r_F + a'_F W_F a_F, \quad C_F = a'_F W_F a_M. \quad (20)$$

The best-response slopes are

$$R_{MF} = -\frac{C_M}{D_M}, \quad R_{FM} = -\frac{C_F}{D_F}. \quad (21)$$

Here R_{MF} is the response of monetary policy to fiscal policy and R_{FM} is the reverse response. The policies are local strategic complements when both cross-curvatures are negative under the present sign convention, so both slopes are positive. Different mandate weights alone can change C_M and C_F , but their effect is quantitatively important only relative to the own curvatures D_M and D_F .

The natural dimensionless interaction index is

$$\Gamma = R_{MF} R_{FM} = \frac{C_M C_F}{D_M D_F}. \quad (22)$$

The inverse of the simultaneous first-order-condition matrix contains $1/(1 - \Gamma)$. Hence Γ measures equilibrium amplification. For $0 < \Gamma < 1$, the complementary best-response map is locally contractive; as $\Gamma \uparrow 1$, shocks and small payoff changes are strongly amplified. At $\Gamma = 1$ the linear system is singular. Negative Γ describes a mixed complement/substitute case rather than mutual complementarity.

There is no universal statistical cut-off, but the following economic classification is useful for the experiments:

Range	Interpretation
$ \Gamma < 0.01$	negligible simultaneous amplification
$0.01 \leq \Gamma < 0.10$	weak but potentially measurable
$0.10 \leq \Gamma < 0.50$	material strategic interaction
$0.50 \leq \Gamma < 0.90$	strong amplification
$0.90 \leq \Gamma < 1$	near-singular coordination problem

These bands are descriptive, not theorem boundaries. The exact stability condition is model-specific, particularly once continuation values enter.

Equation (22) identifies the parameters that matter. Strategic interaction rises when policy transmission a_M, a_F is strong, when both authorities attach material weight to the same affected macro variables, and when direct instrument penalties r_M, r_F are small. Mandate disagreement is neither necessary nor sufficient. If one authority places almost no weight on every outcome moved by the rival, its cross-curvature is close to zero and the product remains weak regardless of the other authority's mandate.

To make the ranges transparent, suppose a common multiplier τ scales both policy-transmission vectors and a common multiplier α scales both instrument penalties. Write $h_j = a'_j W_j a_j$ at the

reference transmission. Then

$$\Gamma(\alpha, \tau) = \frac{\tau^4 C_M C_F}{(\alpha r_M + \tau^2 h_M)(\alpha r_F + \tau^2 h_F)}. \quad (23)$$

This formula generates Figure 1. It is an analytical map, not a re-estimation: the macro curvatures are held at their reference composition while their common scale and the direct policy penalties vary.

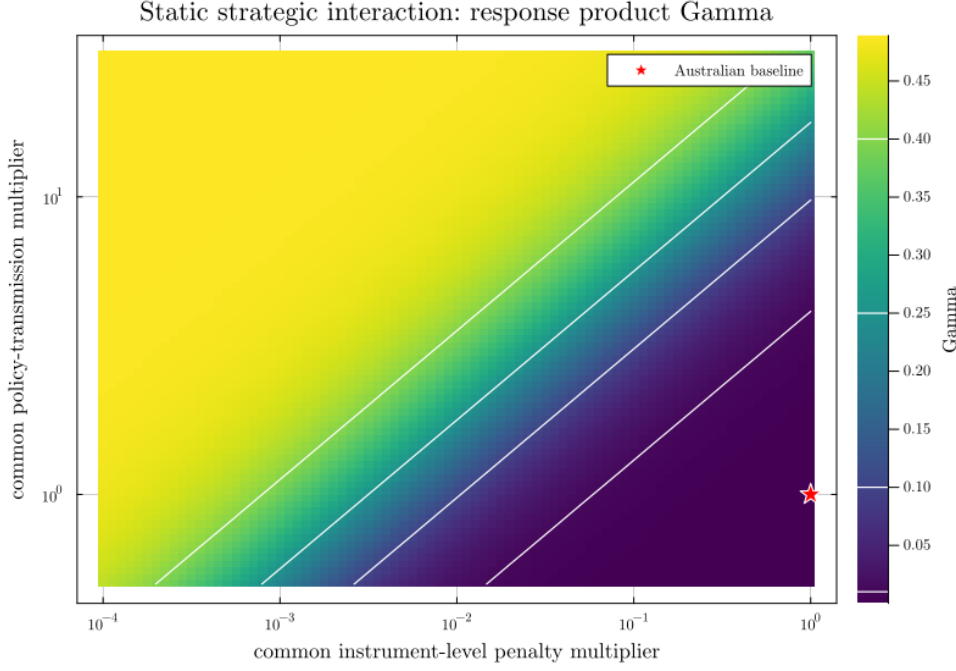


Figure 1: Static response-product ranges before solving the recursive game.

4.2 Dynamic commitment capacity

Adjustment costs should also be normalised. Let D_i denote the one-period own-action curvature before the adjustment term. Define

$$p_i = \frac{\lambda_i}{D_i + \lambda_i} \in [0, 1). \quad (24)$$

In a one-step problem, p_i is the coefficient with which the inherited instrument pulls the current choice back toward its previous value. It is therefore a dimensionless measure of commitment capacity. A raw value such as $\lambda_i = 0.05$ is uninterpretable without D_i .

The limiting cases clarify why the strategic motive is generally hump-shaped. When $p_i = 0$, the inherited instrument is not payoff relevant and the specific lagged-policy commitment channel disappears. When p_i is close to one, the instrument is nearly frozen: the position is durable, but the authority has little ability to move it in response to a shock. Intermediate values give a player both durability and manoeuvrability. The location of the peak is not universal because it depends on discounting, macro persistence, and the rival's feedback rule.

Discounting enters directly. The strategic part of today's first-order condition is proportional to β_i : a myopic authority cannot invest in a future position. Persistent macro and debt states

lengthen the interval over which the rival's response matters. Conversely, a transitory shock can still activate strategic investment, but only through the endogenous policy and debt states that survive it.

An exact local decomposition is available in the solved Markov game. Let K_j be the rival's feedback rule and let $\mathcal{H}_{u_j,t+1}^i$ denote authority i 's total next-period Hamiltonian derivative with respect to the rival's action, including continuation value. The rival-response component of authority i 's current first-order condition is

$$\mathcal{S}_{i,t} = \beta_i(B_i' K_j') \mathcal{H}_{u_j,t+1}^i. \quad (25)$$

The Julia implementation reports the bounded share

$$\Omega_i = \frac{|\mathcal{S}_i|}{|\mathcal{S}_i| + |\mathcal{T}_i - \mathcal{S}_i|}, \quad (26)$$

where $\mathcal{T}_i$ is the entire continuation derivative. Thus Ω_i near zero means that continuation behaviour is mostly ordinary persistence or own smoothing; values above 0.10 indicate a material strategic component, and values above 0.25 a strong one. These last two thresholds are again reporting conventions, not stability conditions.

Figure 2 solves the full recursive game over Γ and p_i . Each panel gives one authority adjustment costs while the other remains flexible. The result makes the two necessary margins visible. Moderate persistence alone does little at the Australian value of Γ . Material strategic shares emerge in parts of the grid once the response product reaches several hundredths and become more systematic around $\Gamma = 0.1$ – 0.4 . The relationship is non-monotone because stronger persistence changes both the rival response and the non-strategic continuation term. The asymmetry across panels reflects the different mandates and debt exposures; it is not imposed by different definitions of the objectives.

A large value of Ω_i is necessary but not sufficient for a large policy distortion. The ratio can rise when the non-strategic continuation term is close to zero, even if the strategic wedge is small in absolute value. A quantitatively important leadership result should therefore satisfy three checks together: material Γ , material Ω_i , and a visible change in equilibrium actions or welfare relative to flexible Nash.

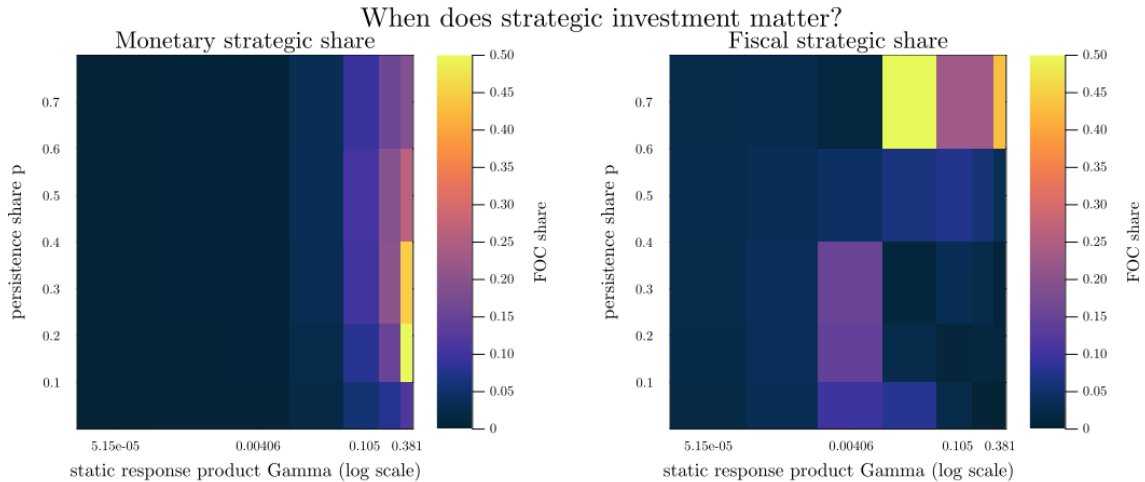


Figure 2: Envelope-based strategic shares over interaction and persistence ranges.

The practical implication is sharp. To generate a Petkov-style leader effect, one should not search over λ_i in isolation. First establish that the static or continuation response product is non-negligible; then vary the dimensionless persistence share; finally verify that (25), rather than the

full continuation term, is large. This sequence prevents ordinary policy inertia from being mistaken for strategic investment.

5 Australian calibration

The quarterly macro coefficients are descriptive OLS estimates for 1993Q1–2019Q4 carried over from the companion Australian data pipeline. The source series are the Reserve Bank of Australia’s public statistical tables ([Reserve Bank of Australia, 2026](#)): consumer prices (G1), GDP and income (H1), demand and income (H2), and interest rates (F1.1). The output gap is constructed from real GDP, the inflation gap from CPI inflation, the monetary instrument from the real cash-rate gap, and the fiscal instrument from public-demand deviations. These are not causal policy multipliers. The debt block, loss weights, and adjustment costs are transparent calibrations.

Table 1: Baseline quarterly calibration

Parameter	Value	Interpretation
ρ_x	0.62875	output-gap persistence
ρ_π	0.56255	inflation persistence
σ	0.03695	monetary transmission to output
ψ	0.05142	fiscal transmission to output
κ	0.15511	Phillips-curve slope
ρ_b	0.995	debt persistence
$(\chi_m, \chi_f, \chi_\pi)$	(0.015, 0.060, 0.100)	debt transmission
(λ_M, λ_F)	(0.050, 0.040)	adjustment-cost primitives
$(\omega_\pi^M, \omega_x^M)$	(1.50, 0.25)	monetary mandate
$(\omega_\pi^F, \omega_x^F)$	(0.50, 0.75)	fiscal mandate
$(\omega_\pi^S, \omega_x^S)$	(1.00, 0.50)	true social objective

The impulse experiments use a one-standard-deviation demand contraction of 0.464 and a cost-push shock of 0.670. Later innovations are zero. Because the model is linear–quadratic, additive shock variances do not change the stationary policy gains under certainty equivalence.

The analytical indices immediately locate this calibration. The static response product is only $5.15e - 05$, well inside the negligible range, whereas the persistence shares are $p_M = 0.332$ and $p_F = 0.281$. Thus commitment capacity is already moderate; weak cross-authority interaction is the binding margin. Holding the composition of macro transmission fixed, reaching $\Gamma = 0.10$ would require both direct instrument penalties to fall to approximately 0.011 of baseline, or both policy-transmission vectors to be about 9.76 times their estimated size. These are large counterfactual movements, not plausible confidence intervals around the OLS coefficients. In the bilateral demand-shock game, the exact bounded strategic shares are only 0.002 for monetary policy and 0.025 for fiscal policy. This establishes analytically and numerically why the baseline mainly displays smoothing rather than endogenous leadership.

6 Results

6.1 Static leadership and dynamic policy positions

Figure 3 is the most direct diagnostic. Diamonds show static Nash, the two Stackelberg orders, and static cooperation. Circles show shock-period actions from the recursive regimes. Monetary persistence mainly reduces the absolute monetary movement; fiscal persistence mainly reduces the fiscal movement. When both costs are present, both adjustments occur.

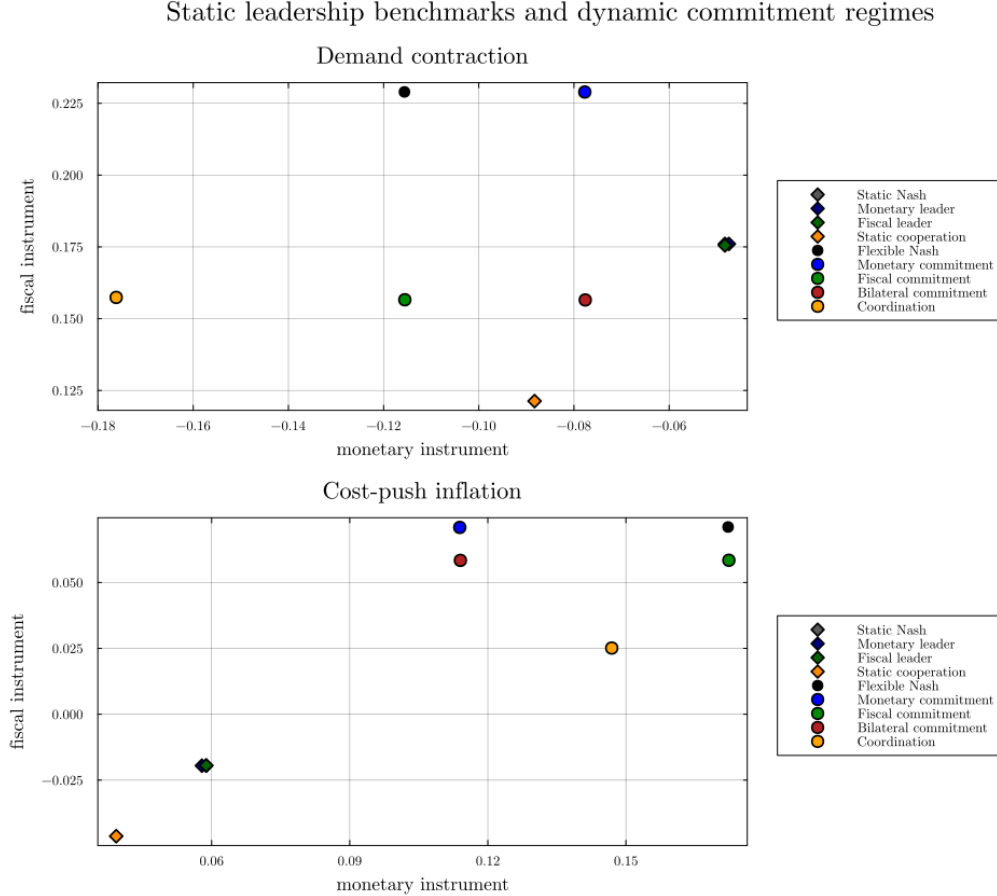


Figure 3: Static policy benchmarks and initial actions from the recursive game.

For the demand shock, monetary persistence closes 0.294 of the flexible-Nash action distance to the monetary leader, while fiscal persistence closes 0.187 toward the fiscal leader. The corresponding cost-push figures are 0.271 and 0.051. These movements are consistent with commitment capacity, but they cannot by themselves identify which authority has become leader: the monetary-leader and fiscal-leader points differ by only 0.0010 after the demand shock and 0.0009 after the cost-push shock. The proximity indices are diagnostics, not structural estimates of leadership.

Figure 4 makes the same warning visible in full feedback-rule space. Fiscal adjustment costs produce the larger shift at the baseline. Because both leader targets are nearly coincident, movement toward one is almost automatically movement toward the other.

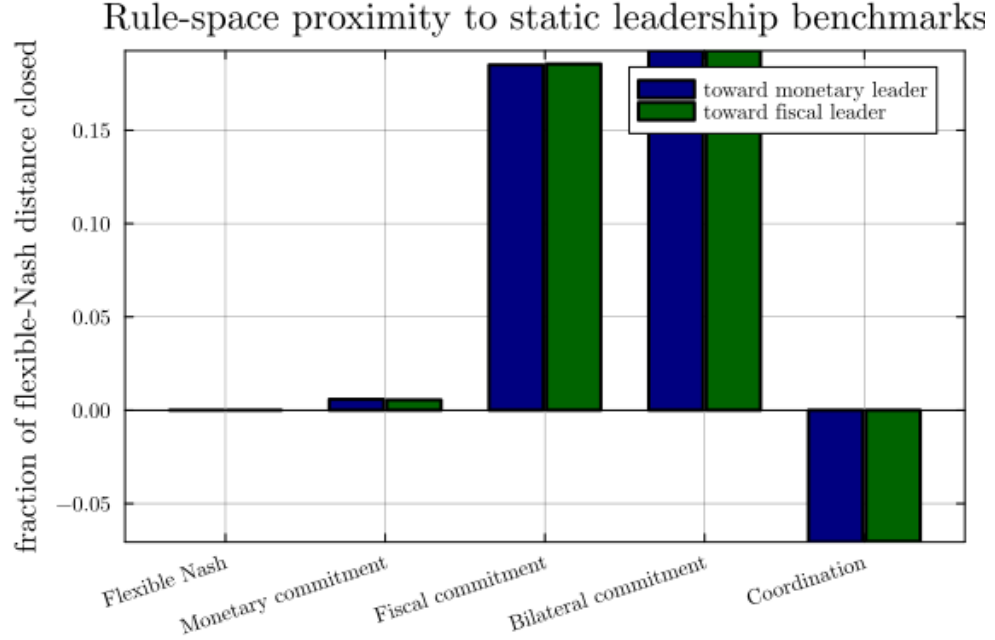
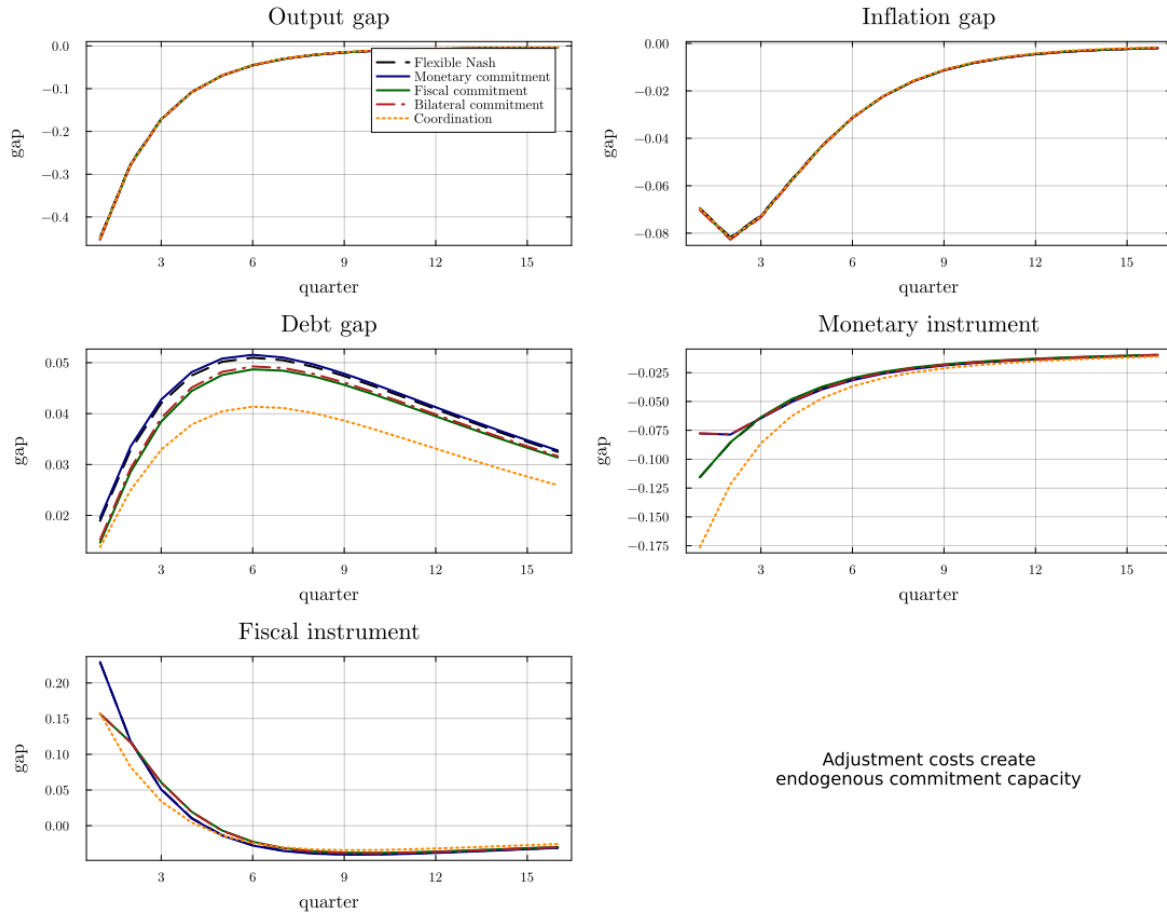


Figure 4: Distance of complete feedback rules from the static benchmarks.

6.2 Shock dynamics and the policy mix

Figures 5 and 6 show 16 quarters of the two impulse responses. Every policy authority reoptimizes in every quarter. The paths are therefore recursive Markov strategies, not commitments to an open-loop sequence. Adjustment costs operate through the inherited policy states in (5).

Australian demand contraction



Adjustment costs create
endogenous commitment capacity

Figure 5: Recursive responses to an Australian demand contraction.

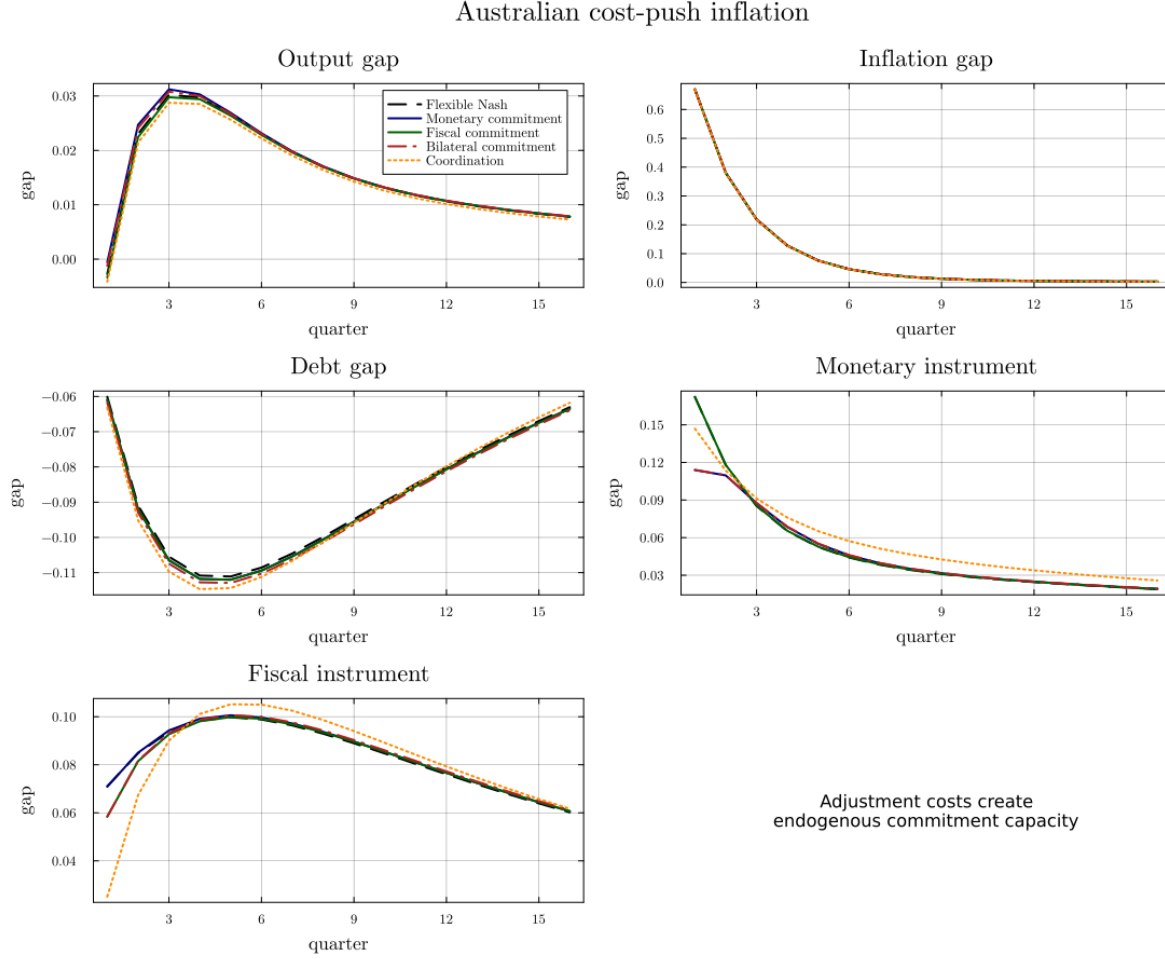


Figure 6: Recursive responses to an Australian cost-push shock.

Table 2 reports the initial policy choices. The flexible game responds sharply because there is no cost to reversing the instruments next quarter. Monetary persistence changes mostly m_0 ; fiscal persistence changes mostly f_0 . Cooperation changes the mix more substantially, especially fiscal policy after a demand contraction. This separates two ideas that were previously conflated: coordination concerns the objective and joint internalisation of spillovers; commitment capacity concerns the state dependence of decentralised play.

Table 2: Initial recursive policy choices

Regime	Demand contraction		Cost-push shock	
	m_0	f_0	m_0	f_0
Flexible Nash	-0.1156	0.2290	0.1722	0.0711
Monetary persistence	-0.0777	0.2289	0.1139	0.0709
Fiscal persistence	-0.1155	0.1566	0.1724	0.0584
Bilateral persistence	-0.0776	0.1565	0.1140	0.0584
Coordination	-0.1762	0.1574	0.1469	0.0251

6.3 Welfare and bilateral commitment capacity

For comparability, Table 3 uses a common macro-welfare yardstick: the true social objective excluding adjustment-resource costs. Actual social loss, including those costs, is also written to the output CSV. The common loss prevents a regime from looking worse merely because a newly introduced physical cost is counted in one column but absent in another.

Table 3: Forty-quarter common macro loss

Regime	Demand contraction	Cost-push shock
Flexible Nash	0.2022	0.7071
Monetary persistence	0.2029	0.7071
Fiscal persistence	0.2017	0.7070
Bilateral persistence	0.2023	0.7070
Coordination	0.2007	0.7066

Monetary persistence slightly raises the common demand-shock loss, while the baseline fiscal persistence slightly lowers it. This is not paradoxical. Smoothing can either impede timely stabilisation or restrain an excessively aggressive delegated response. It is also not a general welfare ranking, because the physical implementation costs have deliberately been omitted from this comparison.

Figure 7 varies one adjustment cost at a time. Fiscal persistence initially brings the feedback rule closer to the static leader region and lowers the macro loss, but the relationship is non-monotone. Monetary persistence ceases to look leader-like at higher costs. A sufficiently large adjustment penalty freezes an instrument; it does not grant unlimited first-mover power.

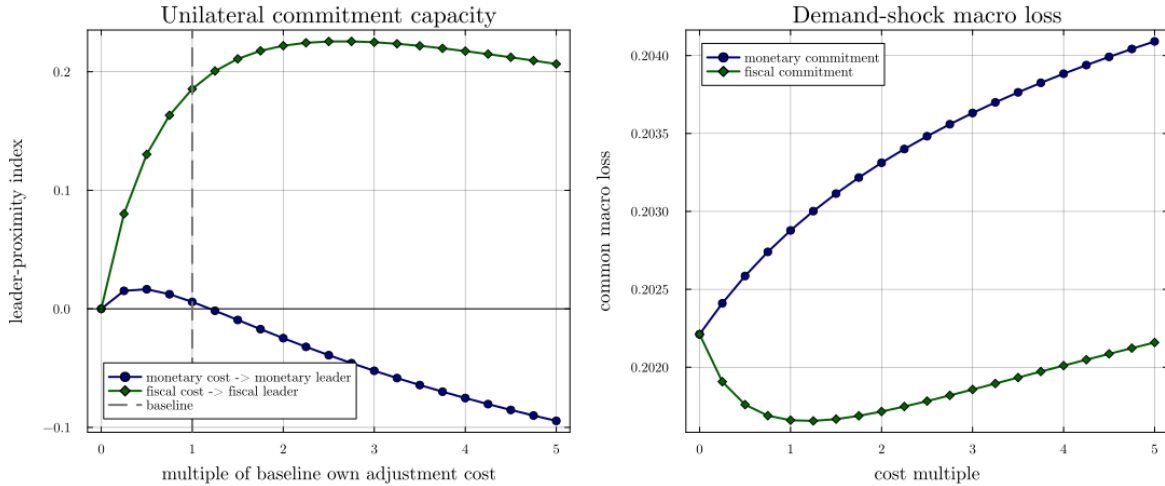


Figure 7: Unilateral commitment capacity and demand-shock loss.

When both capacities vary, Figure 8 shows nearly separable effects on the two initial actions. Figure 9 shows the associated welfare surface. The exercise does not reveal the strong mutual escalation associated with a Stackelberg-warfare point. That absence is consistent with the small follower slopes in (13).

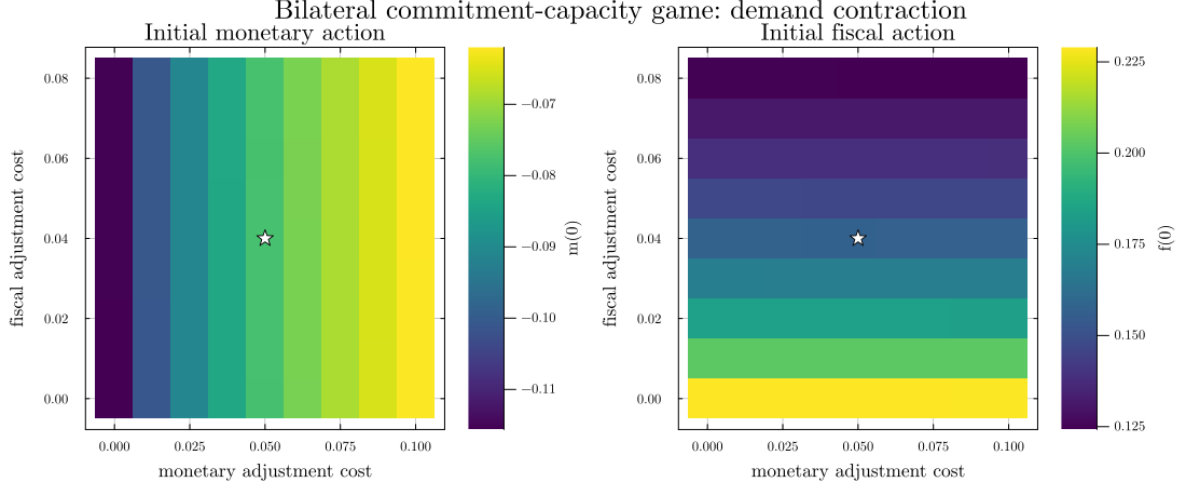


Figure 8: Initial policy choices as both adjustment-cost primitives vary.

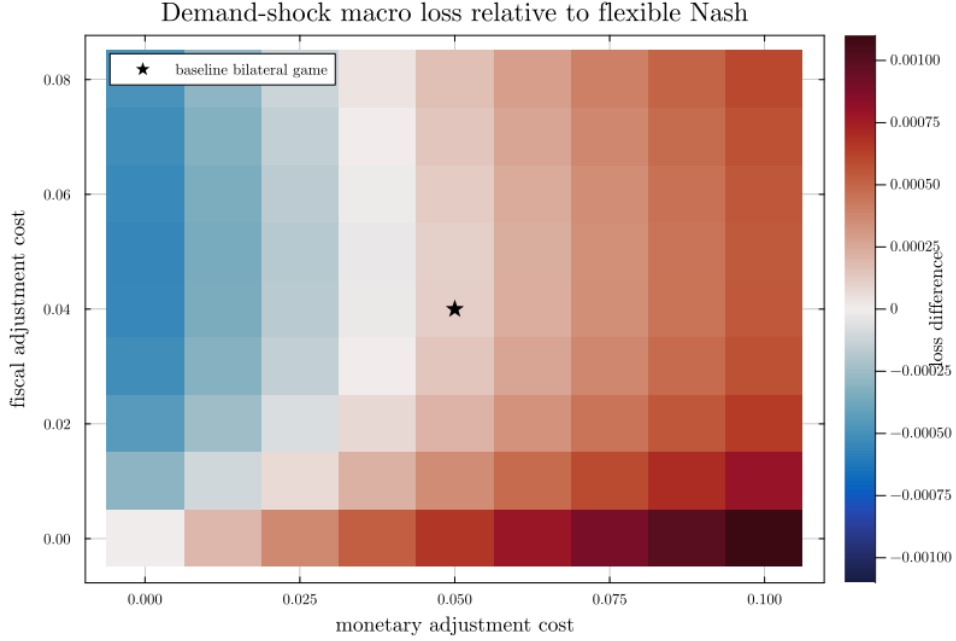


Figure 9: Demand-shock macro loss relative to flexible Nash.

6.4 Stability and equilibrium selection

The bilateral closed-loop spectral radius is 0.9411, below unity. All 6 Riccati initialisations converge to the selected rule; the largest rule distance is 4.99e-11. Thus the numerical search finds one stable stationary linear equilibrium at the baseline. This does not prove global uniqueness. Multiple equilibria would require either additional Riccati roots outside the search basin or nonlinear mechanisms absent from the local quadratic model.

Strategic complementarity and multiplicity must not be equated. Positive best-response slopes can coexist with a unique equilibrium if their product is small. In the present calibration the contemporaneous slopes are not only below the instability threshold but quantitatively close to

zero. The model therefore shows the importance of coordination without yet generating a hard equilibrium-selection problem.

7 Steady state, shocks, and the interpretation of leadership

With zero gaps, zero shocks, and instruments at their neutral values,

$$s^* = 0, \quad m^* = f^* = 0. \quad (27)$$

All quadratic marginal losses vanish there. Adjustment costs also vanish because neither instrument changes. There is consequently no realised strategic-investment wedge at the deterministic steady state, even though the off-equilibrium feedback derivatives remain part of the policy rules.

A shock activates the mechanism. The authorities disagree over the desired mix of output and inflation stabilisation; current controls determine both macro states and inherited instruments; future policymakers respond to those states. Nominal rigidity, represented here by inflation persistence and the Phillips curve, determines how long the disagreement remains payoff relevant. The appropriate solution is therefore a Markov-perfect feedback game, not an open-loop equilibrium.

The current approach captures the strategic-investment mechanism in a narrow, transparent way. It does not yet capture commitment through private-sector expectations. In a fully forward-looking New Keynesian block,

$$\begin{aligned} x_t &= E_t x_{t+1} - \sigma(m_t - E_t \pi_{t+1}) + \psi f_t + d_t, \\ \pi_t &= \beta E_t \pi_{t+1} + \kappa x_t + u_t, \end{aligned}$$

future policy rules change current behaviour directly. That extension could amplify the strategic channel and change the ranking of commitment devices.

8 Forward-looking New Keynesian extension

8.1 Private behaviour and the homotopy

The backward-looking block isolates policy-state investment but omits the effect of expected future policy on private behaviour today. The standard New Keynesian IS curve and Phillips curve make that channel explicit (Clarida, Galí and Gertler, 1999). To distinguish forward-looking behaviour from unrelated recalibration, introduce $\theta \in [0, 1]$:

$$x_t = (1 - \theta)\rho_x x_{t-1} + \theta E_t x_{t+1} - \sigma\{m_t - (1 - \theta)\pi_{t-1} - \theta E_t \pi_{t+1}\} + \psi f_t + d_t, \quad (28)$$

$$\pi_t = (1 - \theta)\rho_\pi \pi_{t-1} + \theta \beta_p E_t \pi_{t+1} + \kappa x_t + u_t. \quad (29)$$

At $\theta = 0$, equations (28)–(29) reproduce the empirical block in (1)–(2). At $\theta = 1$, they become the pure forward-looking NK equations. The debt transition, mandates, adjustment costs, and shock sizes are unchanged. Thus θ is a diagnostic homotopy, not an estimated structural parameter.

Suppose the private sector conjectures

$$\begin{bmatrix} x_t \\ \pi_t \end{bmatrix} = G s_t. \quad (30)$$

Given G , expected future output and inflation are $G s_{t+1}$. Substitution into (28)–(29) yields

$$s_{t+1} = A(G)s_t + B_M(G)m_t + B_F(G)f_t. \quad (31)$$

The authorities solve the simultaneous Markov-perfect game conditional on G . If their rules are $K_M(G)$ and $K_F(G)$ and the contemporaneous private response is $O(G)[s'_t, m_t, f_t]'$, rational expectations require

$$G = O_s(G) + O_m(G)K_M(G) + O_f(G)K_F(G). \quad (32)$$

The Julia solver iterates between the coupled Riccati policy problem and (32), using continuation in θ and damping of expectations. The rule-coefficient error at $\theta = 0$ relative to the original solver is 0.00e+00, an exact implementation check.

Monetary and fiscal authorities move simultaneously with one another. Households and price setters observe current instruments, understand the stationary future rules, and form rational expectations. Future policymakers therefore affect current outcomes through private beliefs, while current policymakers affect future incentives through debt, macro states, and inherited instruments. This is the two-way strategic channel emphasised in work on discretionary equilibrium multiplicity (Kirsanova and Yates, 2025).

8.2 Separating three channels

Forward-looking behaviour requires a refinement of the strategic decomposition. The rival-response term in (25) can be written

$$\mathcal{S}_{i,t} = \beta_i \sum_h B_{i,h} K_{j,h} \mathcal{H}_{u_j,t+1}^i. \quad (33)$$

The implementation separates three objects:

1. the *macro-state channel*, through future x , π , and debt;
2. the *adjustment-state channel*, through m_t becoming m_{t-1} or f_t becoming f_{t-1} next period; and
3. the *private-expectations channel*, through current x_t and π_t depending on the expected future policy rule.

Only the second is the narrow Petkov-style strategic investment created by an instrument-adjustment cost. The first exists with flexible instruments because policy changes future macro states. The third transmits both types of intertemporal strategy from the future back into the present. Calling the complete continuation derivative “strategic investment” would therefore overstate the role of adjustment costs.

8.3 How forward-looking behaviour changes the game

Figure 10 traces demand-shock equilibria. The mapping from θ to policy is strongly non-monotone. As forward-looking behaviour first rises, monetary easing becomes larger and fiscal expansion initially increases. Beyond about $\theta = 0.6$, the fiscal response falls sharply and changes sign. At the pure NK endpoint, flexible Nash chooses $(m_0, f_0) = (-0.0874, -0.0843)$, while bilateral persistence chooses $(-0.0667, -0.0615)$. Fiscal policy contracts after the transitory demand shock: anticipated monetary easing and future stabilisation substitute for a current fiscal expansion, while debt and inflation costs remain.

From backward-looking behaviour to a forward-looking NK economy

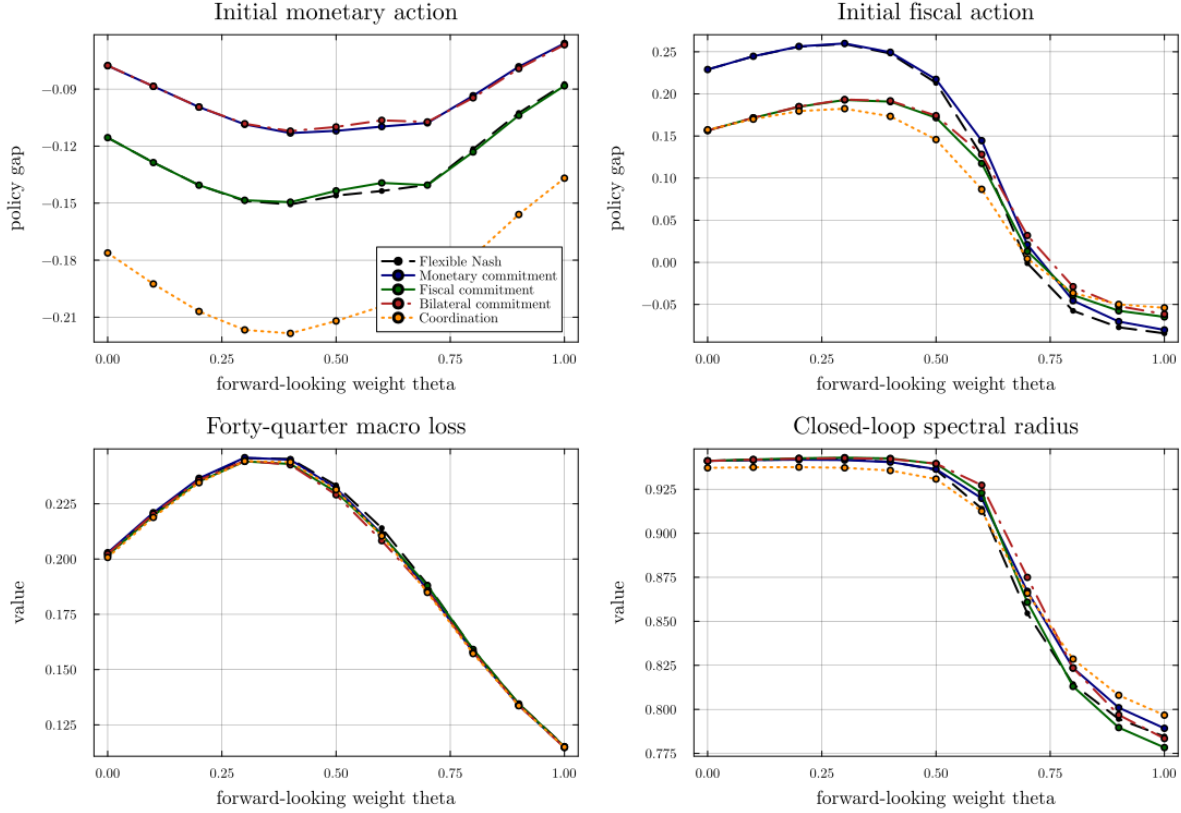


Figure 10: Policy, welfare, and stability along the forward-looking homotopy.

Figure 11 gives the main strategic-investment result. Forward-looking behaviour can amplify the adjustment-state wedge, but the effect is hump-shaped. The monetary wedge peaks around $\theta = 0.80$ at $1.58\text{e-}04$; the fiscal wedge peaks around $\theta = 0.70$ at $1.45\text{e-}04$. At $\theta = 1$, the bounded adjustment-state shares have fallen to 0.003 and 0.002. Private expectations therefore do not mechanically turn instrument smoothing into Stackelberg leadership. They strengthen the channel most when policy positions remain persistent enough to matter but private responses have not already absorbed the future adjustment.

Forward-looking behaviour and strategic investment

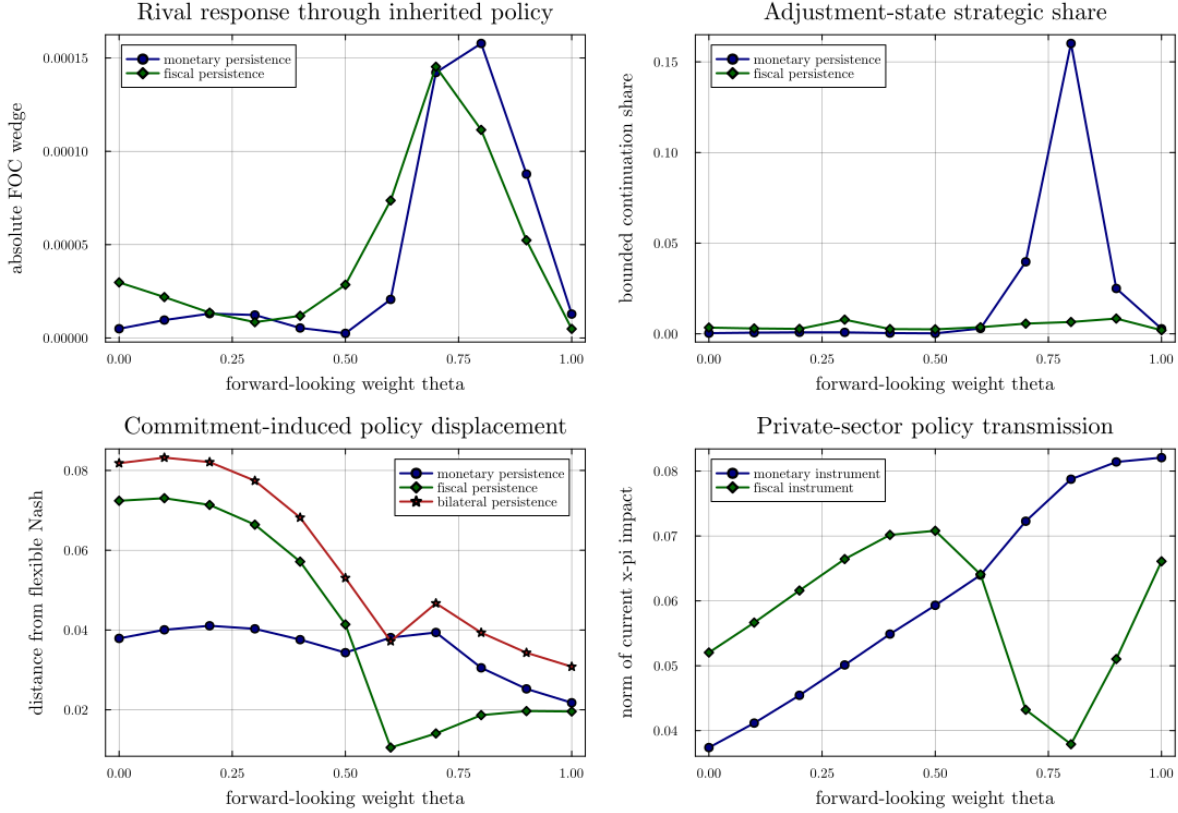
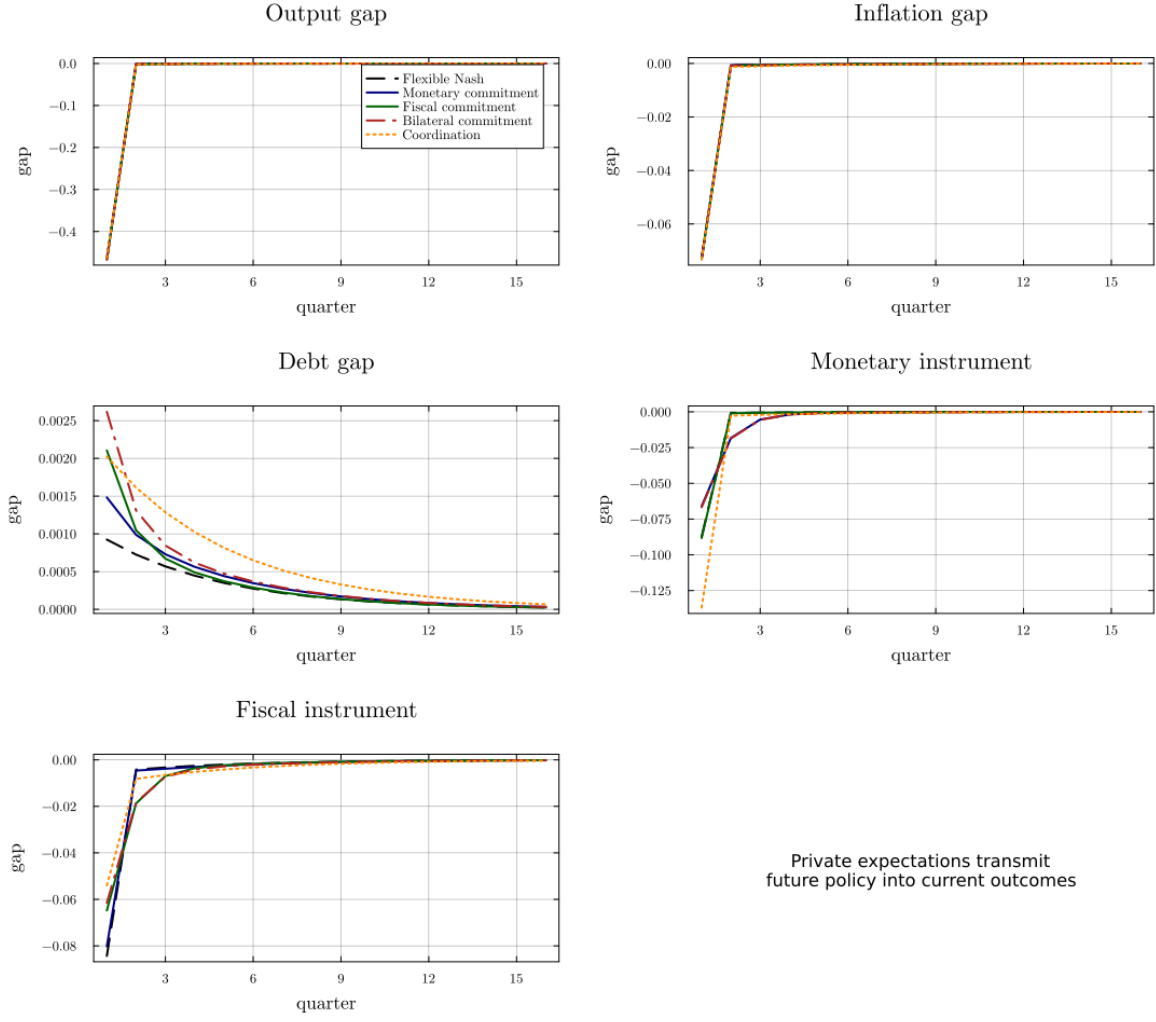


Figure 11: Adjustment-state, macro-state, and private-expectations channels.

The commitment-induced policy displacement becomes smaller at the pure NK endpoint. This is consistent with the general finding that the sign and strength of monetary–fiscal interaction depend on shocks, structural transmission, and policy inertia rather than mandate differences alone (Muscatelli, Tirelli and Trecroci, 2004). Expectations reverse the fiscal sign after a demand disturbance but do not produce an analogous common reversal after a cost-push shock.

The pure-NK impulse responses are shown in Figures 12 and 13. The demand innovation is transitory, so output and inflation move mostly on impact; debt and inherited policies carry its effects forward. The cost-push shock creates more persistent output and debt responses through anticipated stabilisation policy.

Forward-looking NK demand contraction



Private expectations transmit
future policy into current outcomes

Figure 12: Pure forward-looking NK responses to a demand contraction.

Forward-looking NK cost-push inflation

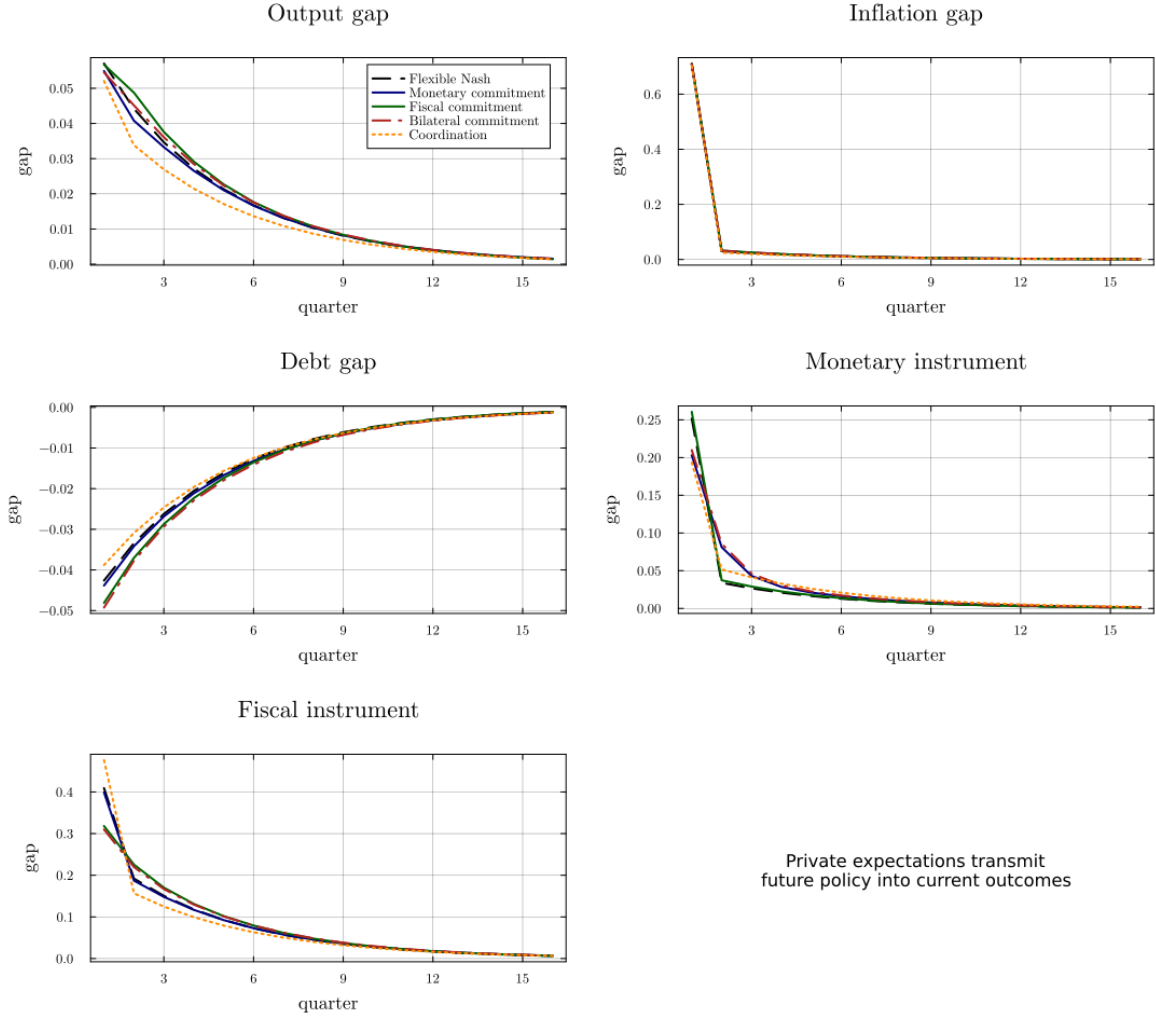


Figure 13: Pure forward-looking NK responses to a cost-push shock.

For the pure-NK demand experiment, common macro loss is 0.1154 under flexible Nash, 0.1147 under bilateral persistence, and 0.1149 under discretionary coordination. The bilateral regime has a slightly lower macro-only loss than the flexible social planner because policy inertia supplies partial commitment. Once physical adjustment costs are included, coordination again has the lowest actual social loss. An institution can improve stabilisation by constraining future discretion while still consuming implementation resources.

8.4 Multiplicity and interpretation

Forward-looking behaviour makes equilibrium selection a genuine concern. Different beliefs about future rules can validate different current private responses and policy choices. [Kirsanova and Yates \(2025\)](#), for example, obtain multiple stable discretionary equilibria in an NK model with capital, where private expectations transmit strategic complementarity between current and future policymakers.

That result is possible, not automatic. Here 7 dispersed initial expectations converge to 1 stable

pure-NK bilateral equilibrium. Closed-loop roots remain inside the unit circle along the homotopy. Expectations materially change the policy mix and create an intermediate strategic-amplification region without generating numerical evidence of multiplicity. The weak response product identified in Section 4 remains a binding limitation.

This is not a global uniqueness theorem. The search covers linear stationary equilibria reachable by the damped fixed-point algorithm. Stronger debt valuation, persistent shocks, lower instrument penalties, capital accumulation, or nonlinear fiscal limits could create additional branches. Those are now well-defined extensions rather than being confounded with the adjustment-cost mechanism itself.

9 Relation to monetary–fiscal and fiscal-dominance literatures

Dynamic monetary–fiscal games distinguish cooperation, discretionary feedback, and commitment (Oudiz and Sachs, 1984; Tabellini, 1986). Public debt is especially important because fiscal choices can change a persistent state to which monetary policy later responds (Beetsma and Bovenberg, 1997). The present debt equation contains that strategic-state channel.

The model does not, however, fully nest the fiscal theory of the price level or strong fiscal dominance. That requires nominal government liabilities, a consolidated intertemporal government budget constraint, and explicit monetary and fiscal backing regimes. Here inflation reduces the debt gap and both authorities respond to debt, so fiscal pressure can influence monetary policy; but the price level is not selected to satisfy a nominal valuation equation. The model should therefore be described as a dynamic policy-coordination game with a fiscal-pressure channel, not as a complete fiscal-dominance model.

The distinction between delegated objectives and social welfare is also standard in monetary economics. Rogoff-style conservatism deliberately gives the central bank a different inflation weight from society, while the social objective remains fixed. In the present application, the fiscal authority is relatively output-oriented and monetary policy relatively inflation-oriented. Those different weights create conflict, but conflict alone need not create a large leader advantage. Transmission and reaction slopes determine whether one authority can use persistence to manipulate the other.

10 Conclusion

The requested dynamic game is a recursive, simultaneous policy game with lagged instruments as endogenous states. It differs from the original static Stackelberg game, while retaining the latter as a benchmark. With only one adjustment cost, that authority possesses unilateral commitment capacity; with both, both authorities try to shape the state inherited by future policymakers. The model therefore implements the strategic-investment logic discussed by Petkov and by Jun and Vives.

The Australian baseline does not generate a strong endogenous leader. The reason is economic rather than computational: the static follower responses are weak, and the two leader allocations are nearly coincident. Adjustment costs mainly generate own-instrument inertia. Cooperation still matters for the policy mix, but the stationary linear game has one numerically detected stable equilibrium rather than an equilibrium-selection problem.

Forward-looking private behaviour changes the policy mix substantially and temporarily amplifies the adjustment-state wedge at intermediate values of the homotopy. It does not monotonically magnify strategic investment, and the pure NK endpoint again has small adjustment-specific shares

and one detected stable equilibrium. Expectations are a conduit, not an independent guarantee of leadership or multiplicity.

The next useful experiment is therefore not to raise adjustment costs indiscriminately. It is to strengthen and estimate the economically relevant cross-effects: policy transmission, debt valuation, shock persistence, and mandate divergence. Adding capital or nominal government liabilities would then test whether a stronger endogenous state creates additional rational-expectations equilibrium branches.

A Reproduction

All model calculations and figures are generated in Julia. From this folder:

```
julia --project=. -e "using Pkg; Pkg.instantiate(; update_registry=false)"
julia --project=. run_model.jl
julia --project=. run_nk_extension.jl
julia --project=. runtests.jl
julia --project=. runtests_nk.jl
julia --project=. build_tex.jl
```

The first model command writes calibration, policy rules, equilibrium-search diagnostics, summary tables, and figures to `output/`. The final command uses the artifact-pinned Tectonic binary to compile this source into `STRATEGIC_INVESTMENT_POLICY_GAME.pdf`.

B What the numerical checks establish

The test suite verifies the static Nash first-order conditions, each Stackelberg follower condition, ownership of the adjustment costs, convergence and stability of every stationary regime, irrelevance of inherited instrument states when adjustment costs are zero, activation of those states when costs are positive, multi-start agreement, and finite simulated losses. These tests establish internal numerical consistency. They do not establish structural identification of the calibrated weights or global equilibrium uniqueness. The NK suite additionally verifies the exact $\theta = 0$ replication, both forward-looking private equations at an arbitrary state, the rational-expectations fixed point, the channel decomposition, and convergence from dispersed initial beliefs.

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